

Disc tilt approximation for the flapping motion

Flapping equation – hovering flight

$$\ddot{\beta} + \frac{\gamma}{8}\Omega\dot{\beta} + \Omega^2\beta = \frac{\gamma}{8}\Omega^2\left[\theta_0 - \frac{4\lambda_1}{3}\right]$$

inertia aero damping centrifugal “spring” external excitation (const)

 Rotating system

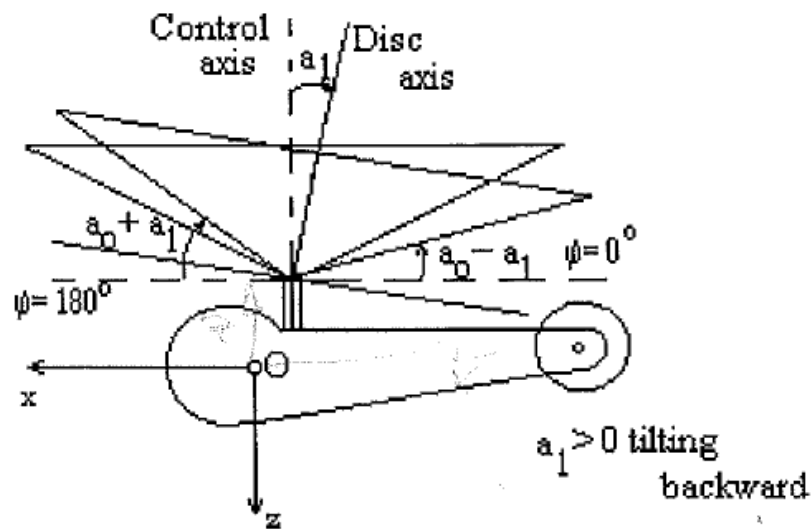
Disc tilt approximation for flapping motion

There is a disc approach to study the flapping motion obtained by a Fourier series applied to the flapping motion to transform it from rotating to the non-rotating plane

$$\beta = a_0 - a_1 \cos \psi - b_1 \sin \psi$$

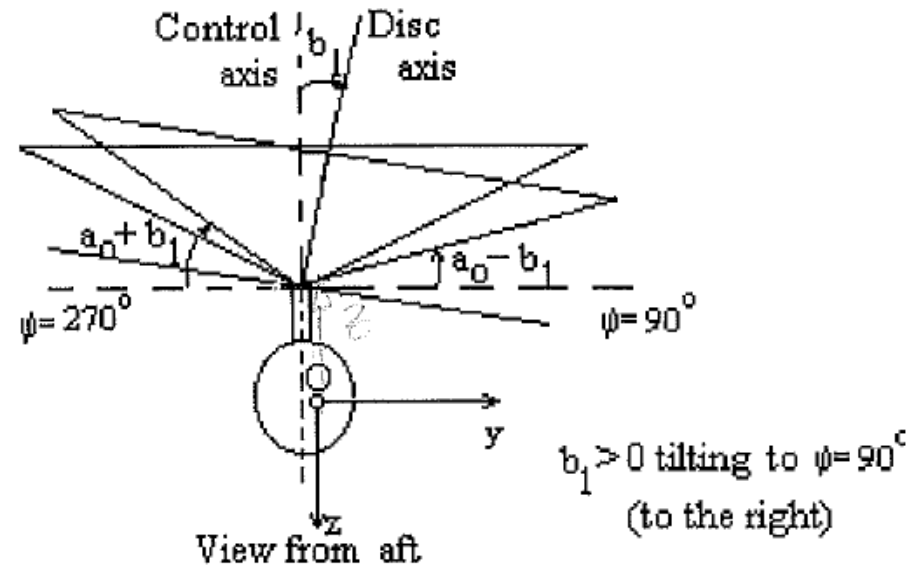
The diagram shows the equation $\beta = a_0 - a_1 \cos \psi - b_1 \sin \psi$ inside a light blue rectangular box. Three arrows point from labels below to terms in the equation: an arrow from 'coning angle' points to a_0 , an arrow from 'longitudinal disc tilt' points to a_1 , and an arrow from 'lateral disc tilt' points to b_1 .

Interpretation of disc tilt



$$\psi = 0^\circ \rightarrow \beta = a_0 - a_1$$

$$\psi = 180^\circ \rightarrow \beta = a_0 + a_1$$



$$\psi = 90^\circ \rightarrow \beta = a_0 - b_1$$

$$\psi = 270^\circ \rightarrow \beta = a_0 + b_1$$

Disc tilt approximation for flapping motion

Assume no flapping dynamics for the disc:

$$\psi = \Omega t$$

$$\beta = a_0 - a_1 \cos \psi - b_1 \sin \psi$$

$$\dot{\beta} = \Omega (a_1 \sin \psi - b_1 \cos \psi)$$

$$\ddot{\beta} = \Omega^2 (a_1 \cos \psi + b_1 \sin \psi)$$

Substituting this into flapping equation for hover and collecting the free term, the cosine and sine terms gives:

Disc tilt approximation for flapping motion

$$(a_1 \cos \psi + b_1 \sin \psi) + \frac{\gamma}{8}(a_1 \sin \psi - b_1 \cos \psi) +$$

$$(a_0 - a_1 \cos \psi - b_1 \sin \psi) = \frac{\gamma}{8} \left[\theta_0 - \frac{4\lambda_1}{3} \right]$$

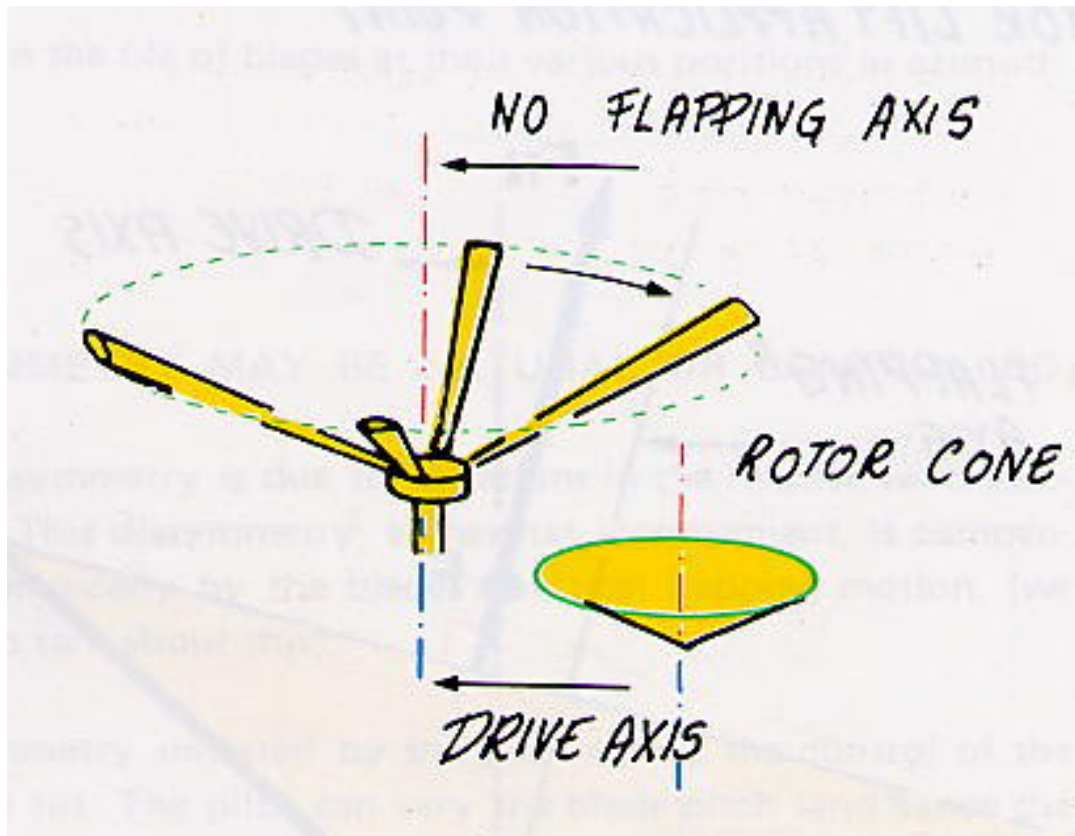
Equalizing each term results in 3 equations:

$$a_0 = \frac{\gamma}{8} \left[\theta_0 - \frac{4\lambda_1}{3} \right]$$

$$\left(a_1 - \frac{\gamma}{8} b_1 - a_1 \right) \cos \psi = 0$$

$$\left(b_1 + \frac{\gamma}{8} a_1 - b_1 \right) \sin \psi = 0$$

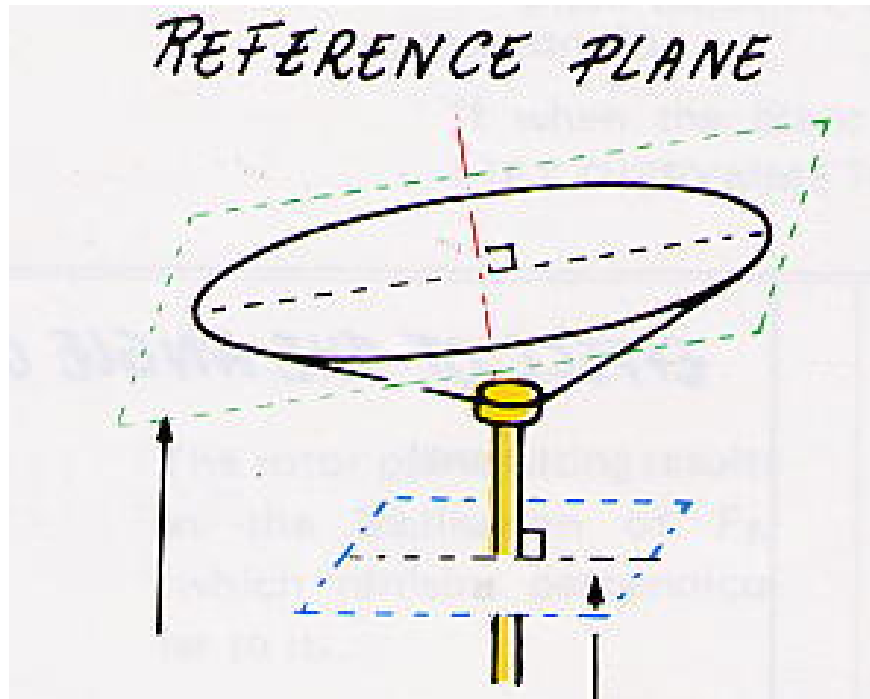
Coning angle



Coning angle

$$a_0 = \frac{\gamma}{8} \left[\theta_0 - \frac{4\lambda_1}{3} \right]$$

Flapping motion – nonrotating system



a_1 : longitudinal tilt of the rotor cone
 b_1 : lateral tilt of the rotor cone

In hovering flight longitudinal and lateral tilt are zero.